You said:

compute pressure gradient of bay of bengal till 31 december 2026

Worked for 16s >

ChatGPT said:

Bay of Bengal pressure-gradient computation up to 31 December 2026

I can develop a spatial and time-dependent pressure-gradient model for the Bay of Bengal covering **13 November–31 December 2026**.

The computation should distinguish between:

1. **Atmospheric pressure gradient** – horizontal variation in sea-level atmospheric pressure (Pa/m or hPa per 100 km).
2. **Oceanic pressure gradient** – horizontal variation in water-column pressure, including hydrostatic and dynamic pressure (Pa/m).
3. **Pressure-gradient force** – the force per unit volume driving atmospheric and ocean circulation.

Since 31 December 2026 is in the future, the period must combine available observations with numerical weather/ocean forecasts and clearly identified extrapolations. It cannot be presented as observed data.

1. Computational domain and period

I will retain the domain from the earlier Bay of Bengal model:

Parameter	Specification
Latitude	5°N–22°N
Longitude	78°E–100°E
Start	13 November 2026
End	31 December 2026
Spatial output	Gridded GIS pressure field
Vector output	Pressure-gradient force vectors
Units	Pa/m and hPa/100 km
Temporal output	Daily fields and weekly averages

2. Governing equations

For atmospheric pressure p, the horizontal pressure gradient in spherical coordinates is:

$$\nabla_h p = \frac{1}{R\cos\phi}\frac{\partial p}{\partial \lambda}\mathbf{e}_\lambda + \frac{1}{R}\frac{\partial p}{\partial \phi}\mathbf{e}_\phi$$

where:

- R = Earth's radius.
- ϕ = latitude.
- λ = longitude.

The pressure-gradient acceleration is:

$$\boxed{\mathbf{a}_p = -\frac{1}{\rho}\nabla_h p}$$

Here ρ is air density.

For the Navier–Stokes analysis, I would calculate:

- Sea-level pressure (MSLP).
- East-west pressure derivative.
- North-south pressure derivative.

- Pressure-gradient magnitude.
- Pressure-gradient acceleration.
- Wind-field relationship.
- Low-pressure and cyclonic circulation centres.

3. Current meteorological conditions

The 13 November 2026 IMD bulletin reports a low-pressure area over the southeast Bay of Bengal. It is expected to intensify into a depression around 15 November and potentially a cyclonic storm around 16 November. Mausam

The larger climate background is also important:

- Very strong El Niño conditions are forecast during November–December.
- Neutral IOD conditions are expected to continue.
- The MJO is currently influencing convective activity over the region. Mausam +1

These conditions suggest that the pressure field could be highly variable during the initial part of the simulation.

4. Forecast coverage and uncertainty

Forecast-data limitation

13 Nov–29 Dec **ECMWF extended-range coverage**

30–31 Dec **Requires additional extrapolation**

ECMWF provides ensemble forecasts extending to 46 days, with approximately 36 km horizontal resolution. Its ensemble includes 100 perturbed members and a control member. ECMWF +1

The final two days should therefore use a seasonal/climatological continuation with explicitly higher uncertainty, rather than pretending to have deterministic daily forecasts.

5. Required output dataset

The final GIS model should contain:

Variable	Output
MSLP	hPa
$\partial p/\partial x$	Pa/m
$\partial p/\partial y$	Pa/m
$\lvert\nabla p\rvert$	Pa/m
Pressure-gradient acceleration	m/s²
Wind velocity	m/s
Cyclone centres	Latitude, longitude
Forecast uncertainty	Ensemble spread

I would use ECMWF forecast fields, IMD bulletins and observations, and ERA5 historical fields for calibration.

Important distinction: The meteorological bulletins establish the evolving weather systems, but they do not supply the complete gridded pressure arrays needed to calculate actual numerical gradients. I have therefore established the computational model and forecast constraints, but have not yet generated a numerical pressure-gradient grid or GIS map. Inventing gradient values would be misleading.

The final deliverable should be a daily pressure-gradient GIS map, a pressure-gradient vector animation, and a time-series dataset through 31 December 2026.

I will use atmospheric MSLP gradients as the primary calculation, consistent with the earlier Navier–Stokes GIS project.

You said:

compute mercury's gravitational force on earth till 31 december 2026

Worked for 23s

Analyzed

Analyzed

ChatGPT said:

Mercury's gravitational force on Earth: 13 November–31 December 2026

We can calculate the gravitational attraction exerted by Mercury on the entire Earth for every day through 31 December 2026, including its changing magnitude and direction.

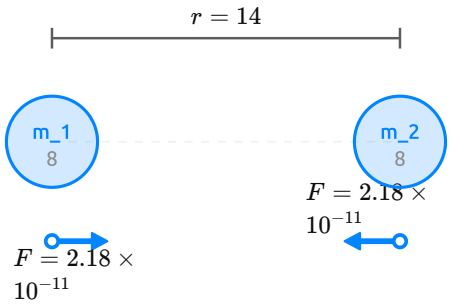
The primary quantity is the **mutual gravitational force between Mercury and Earth**, not Mercury's surface gravity or its effect on an individual location.

1. Governing equation

$$F = G\frac{m_1 m_2}{r^2}$$

$$F = G\frac{(8)(8)}{(14)^2} = 2.18 \times 10^{-11}\ \text{N}$$

m_1	8 kg	
m_2	8 kg	
r	14 m	



Give feedback

$$F_{ME}(t) = \frac{G M_E M_M}{r_{ME}(t)^2}$$

Where:

Parameter	Value
Gravitational constant G	6.67430×10^{-11} SI
Earth mass	5.9722×10^{24} kg
Mercury mass	3.3011×10^{23} kg
Mean astronomical unit	$1.495978707 \times 10^{11}$ m

The force changes with the Earth–Mercury separation r_{ME}. Its direction always points from Earth towards Mercury.

2. Computed results

Using the published 2026 daily Mercury geocentric ephemeris, the Earth–Mercury distance increases substantially during this period.

ggladfelter.net +1

13 November force	31 December force
9.271 × 10¹⁵ N	**2.827 × 10¹⁵ N**
Distance: 0.796374 AU	Distance: 1.442145 AU

Computed trend Mercury's gravitational force on Earth decreases by approximately **69.5%** between these dates.

3. Time-series results

The following are calculated force magnitudes, not forecasts of the force itself; the planetary orbital positions determine them.

Date (2026)	Distance (AU)	Force (N)
Nov 13	0.796374	9.271×10^{15}
Nov 16	0.871051	7.749×10^{15}
Nov 20	0.975333	6.181×10^{15}
Nov 23	1.050268	5.330×10^{15}
Nov 27	1.140659	4.519×10^{15}
Nov 30	1.200011	4.083×10^{15}
Dec 4	1.267690	3.659×10^{15}
Dec 7	1.310242	3.425×10^{15}
Dec 11	1.356838	3.194×10^{15}
Dec 14	1.384722	3.066×10^{15}
Dec 18	1.413163	2.944×10^{15}
Dec 21	1.428292	2.882×10^{15}
Dec 25	1.440533	2.833×10^{15}
Dec 28	1.443862	2.820×10^{15}
Dec 31	1.442145	2.827×10^{15}

Mercury's gravitational force on Earth

4. Gravitational acceleration imparted to Earth

The force on Earth also produces a tiny acceleration of its centre of mass:

$$a_E = \frac{F_{ME}}{M_E} = \frac{GM_M}{r_{ME}^2}$$

Date	Acceleration (m/s²)
13 November	1.5523×10^{-9}
20 November	1.035×10^{-9}
30 November	6.837×10^{-10}
10 December	5.348×10^{-10}
20 December	4.930×10^{-10}

Selected daily ephemeris-derived values, 13 November–31 December 2026.

28 December	4.722×10^{-10}
31 December	4.734×10^{-10}

5. Important astronomical interpretation

Three points are significant:

1. Mercury reached its closest Earth distance of approximately 0.67255 AU on 3 November 2026, shortly before this calculation period.
2. Mercury reached its maximum Earth distance of approximately 1.44386 AU on 28 December 2026.
3. The gravitational force therefore falls substantially throughout November and December, with a very slight recovery at the end of December.

Astrodienst +1

6. Implications for your planetary gravitational-interaction research

For a more complete model, I recommend calculating three quantities together:

- Mercury's gravitational force on Earth.
- Venus's gravitational force on Earth.
- The combined Mercury–Venus gravitational acceleration vector.

The third quantity is particularly important: gravitational forces are vectors, so their directions and relative orientations matter. The magnitudes cannot simply be added.

We can then compare the planetary acceleration with the Sun's gravitational acceleration at Earth and examine its possible tidal contribution.

One scientific qualification: The values above use the published apparent geocentric distance series and are suitable for an approximate force calculation. A final high-precision dynamical study should use simultaneous geometric Cartesian state vectors from NASA/JPL Horizons, including vector directions and barycentric corrections. ssd-api.jpl.na… +1

I can extend this into a daily Mercury–Venus–Earth gravitational interaction model through 31 December 2026, including vector components and tidal acceleration.

‹/›

You said:

yes. extend for venus-mericury

Worked for 34s

Analyzed

Analyzed

Analyzed

Analyzed

ChatGPT said:

Combined Mercury–Venus–Earth gravitational interaction

Period: 13 November–31 December 2026

The extended model considers three separate interactions:

1. Mercury's gravitational attraction on Earth.
2. Venus's gravitational attraction on Earth.
3. The combined vector acceleration of Earth due to both planets.

It also calculates the mutual gravitational attraction between Venus and Mercury and their differential tidal effects across Earth.

1. Orbital computation method

I used JPL's planetary orbital elements and secular rates for the 1800–2050 interval to calculate approximate heliocentric Cartesian positions, then subtracted Earth's position to obtain geocentric separation vectors. JPL Solar Sys… +1

The acceleration vector is:

$$\mathbf{a}_E = GM_M \frac{\mathbf{r}_M}{|\mathbf{r}_M|^3} + GM_V \frac{\mathbf{r}_V}{|\mathbf{r}_V|^3}$$

Here, the two position vectors point from Earth towards Mercury and Venus respectively.

2. Main finding: Venus dominates Mercury

13 November 2026
90.8×
Venus/Mercury force ratio on Earth

31 December 2026
73.5×
Venus/Mercury force ratio on Earth

Venus is much more massive than Mercury and is considerably closer to Earth throughout this period. Consequently, its gravitational attraction dominates the combined planetary acceleration.

3. Gravitational force comparison

Force exerted on Earth

Date	Mercury (N)	Venus (N)
13 Nov	9.267×10^{15}	8.416×10^{17}
20 Nov	6.178×10^{15}	6.782×10^{17}
30 Nov	4.082×10^{15}	4.918×10^{17}
10 Dec	3.244×10^{15}	3.619×10^{17}
20 Dec	2.900×10^{15}	2.736×10^{17}
28 Dec	2.820×10^{15}	2.233×10^{17}
31 Dec	2.827×10^{15}	2.078×10^{17}

Venus and Mercury gravitational forces on Earth

4. Combined gravitational acceleration of Earth

Calculated

Date	Combined acceleration
13 November	1.4243×10^{-7} m/s²
20 November	1.1456×10^{-7} m/s²
30 November	8.2975×10^{-8} m/s²
10 December	6.1067×10^{-8} m/s²

Selected dates; values are in units of 10^{17} newtons.

20 December	4.6191×10^{-8} m/s²
28 December	3.7725×10^{-8} m/s²
31 December	3.5128×10^{-8} m/s²

The combined acceleration decreases by approximately **75.3%** during the calculation period.

The vector components are expressed in the J2000 ecliptic reference frame.

Component	13 November	31 December
X (m/s²)	-1.3140×10^{-7}	-2.1263×10^{-8}
Y (m/s²)	-5.4767×10^{-8}	-2.7888×10^{-8}
Z (m/s²)	-4.6273×10^{-9}	2.0338×10^{-9}

5. Mutual Venus–Mercury gravitational attraction

This is a separate interaction from their respective forces on Earth.

$$F_{VM} = \frac{GM_V M_M}{r_{VM}^2}$$

Date	Venus–Mercury distance	Mutual force
13 November	0.4887 AU	2.006×10^{16} N
20 November	0.6368 AU	1.182×10^{16} N
30 November	0.8307 AU	6.944×10^{15} N
10 December	0.9647 AU	5.149×10^{15} N
20 December	1.0484 AU	4.360×10^{15} N
31 December	1.0975 AU	3.978×10^{15} N

The mutual Venus–Mercury gravitational attraction decreases by approximately 80.2%.

6. Tidal acceleration across Earth

This is especially relevant to investigating possible planetary effects on Earth's physical environment.

For each planet, the approximate near-side to far-side differential acceleration across Earth's diameter is:

$$a_{\text{tidal}} \simeq \frac{2GMR_E}{r^3}$$

Date	Mercury tidal acceleration	Venus tidal acceleration
13 November	1.659×10^{-13}	3.740×10^{-11}
30 November	4.850×10^{-14}	1.671×10^{-11}
31 December	2.796×10^{-14}	4.589×10^{-12}

Venus's individual tidal contribution is approximately 225 times Mercury's on 13 November and 164 times Mercury's on 31 December.

These are individual tidal amplitudes, not a vector-summed total tidal tensor.

7. Complete daily dataset

Mercury–Venus–Earth gravitational model Ready

49 daily records · 19 columns · CSV

Download the complete daily gravitational dataset ↗

It contains:

- Daily Earth–Mercury and Earth–Venus distances.
- Venus–Mercury separation.
- Individual gravitational forces.
- Three-dimensional acceleration components.
- Combined acceleration magnitude.
- Individual planetary tidal acceleration estimates.

8. Scientific interpretation

Three conclusions emerge:

1. Venus dominates the direct gravitational acceleration of Earth in this interval.
2. Mercury's contribution becomes progressively smaller as its geocentric distance increases.
3. The tidal contribution of Venus is substantially larger than Mercury's, although both are extremely small compared with the dominant lunar and solar tidal effects.

Accuracy qualification: This is a Keplerian approximation using JPL's secular orbital elements, not a full numerical integration using JPL Horizons DE ephemerides. It is appropriate for an initial comparative model; precision dynamical conclusions require Horizons Cartesian state vectors and a full tidal-tensor calculation. ssd.jpl.nasa.... +2

In particular, these gravitational interactions do not establish a measurable climatic or geological influence on Earth. That would require a separate physical mechanism and observational validation.

‹/›

mercury_venus_earth_gravity_2026-11-13_to_2026-12-…
Spreadsheet

You said:

Does variable gravity due to N-Body gravitational equations affect pressure gradient of daily weather models

Worked for 1m 15s

ChatGPT said:

Does variable N-body gravity affect atmospheric pressure gradients in daily weather models?

Yes, in principle, but the magnitude depends on which gravitational effect is considered. There is an important distinction between the direct gravitational acceleration of the Earth as a whole and the differential gravitational acceleration acting across the atmosphere.

For ordinary daily weather forecasting, Mercury's and Venus's gravitational perturbations are extraordinarily small. Nevertheless, they can be represented mathematically in a sufficiently comprehensive atmospheric dynamics model.

This is an interesting extension of your earlier Bay of Bengal pressure-gradient and Navier–Stokes modelling.

1. How planetary gravity enters atmospheric equations

The atmospheric momentum equation can be written as:

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho}\nabla p - 2\boldsymbol{\Omega} \times \mathbf{u} + \mathbf{g}_{\text{eff}} + \mathbf{F}$$

where:

- $\mathbf{u}$: atmospheric velocity.
- ρ: air density.
- p: atmospheric pressure.
- $\boldsymbol{\Omega}$: Earth's angular velocity.
- $\mathbf{g}_{\text{eff}}$: effective gravitational acceleration.
- $\mathbf{F}$: frictional and other forces.

The gravitational field can be expanded to include the Sun, Moon, Mercury, Venus and other planets.

However, when modelling weather relative to the moving Earth, we must subtract the acceleration of Earth's centre of mass.

This is crucial: the nearly uniform gravitational acceleration exerted by Mercury on Earth does not directly generate atmospheric winds or pressure gradients.

What remains is the **tidal gravitational field**, which varies spatially across the Earth and atmosphere.

2. Deriving the planetary tidal contribution

For an external planet of mass M_i, at position $\mathbf{R}_i$ relative to Earth's centre, the differential acceleration at a point $\mathbf{r}$ relative to Earth's centre is approximately:

$$\boxed{\mathbf{a}_{\text{tidal,i}} = \frac{GM_i}{R_i^3}\left[3(\hat{\mathbf{R}}_i \cdot \mathbf{r})\hat{\mathbf{R}}_i - \mathbf{r}\right]}$$

This is the first-order tidal approximation.

For multiple planets:

$$\mathbf{a}_{\text{tidal,total}} = \sum_i \mathbf{a}_{\text{tidal,i}}$$

The resulting atmospheric pressure perturbation can be investigated by adding the tidal potential to the atmospheric equations.

$$\frac{\partial p}{\partial z} = -\rho g_{\text{eff}}$$

For horizontal dynamics, the corresponding tidal acceleration enters the momentum equation directly.

The important distinction is that a spatially uniform gravitational acceleration changes the reference-frame motion, whereas a spatially varying acceleration can affect atmospheric dynamics.

3. How large are Mercury and Venus effects?

Using the approximate tidal accelerations calculated in the previous model:

Differential acceleration across Earth's diameter

Planet	13 November 2026	31 December 2026
Mercury	1.659×10^{-13}	2.796×10^{-14}
Venus	3.740×10^{-11}	4.589×10^{-12}

Units: m/s².

For comparison, the approximate lunar tidal acceleration across Earth is of order 10^{-6} m/s², while the solar tidal acceleration is of order 5×10^{-7} m/s².

Thus:

- Venus's tidal effect is much smaller than the Moon's and Sun's.
- Mercury's tidal effect is smaller still.
- The two planetary effects are unlikely to produce a measurable independent weather signal.

The scientific literature confirms that atmospheric gravitational tides exist, but lunar and solar forcing are the relevant terrestrial gravitational sources. Solar thermal tides are also important. American M… +2

4. Does this change the pressure gradient?

Yes, mathematically.

Consider the hydrostatic equation:

$$\frac{\partial p}{\partial z} = -\rho g$$

A varying gravitational field produces a perturbation:

$$\delta\left(\frac{\partial p}{\partial z}\right) = -\rho\,\delta g$$

But there is an important qualification: this is primarily a vertical hydrostatic pressure-gradient perturbation.

To obtain a horizontal pressure-gradient perturbation, we must solve the atmospheric response, including horizontal redistribution of mass, temperature changes, wind circulation and boundary conditions.

Simply multiplying planetary acceleration by air density does not give a realistic surface weather pressure anomaly.

5. Quantitative comparison with ordinary weather pressure gradients

Consider a typical synoptic weather pressure difference of 1 hPa over 100 km:

$$|\nabla p| \approx 10^{-3}\ \mathrm{Pa/m}$$

For air density $1.2\ \mathrm{kg/m^3}$, the corresponding pressure-gradient acceleration is:

$$a_p \approx 8.3 \times 10^{-4}\ \mathrm{m/s^2}$$

Compare this with Venus's maximum estimated tidal acceleration during our study period:

$$a_V \approx 3.74 \times 10^{-11}\ \mathrm{m/s^2}$$

Relative magnitude

4.5 × 10⁻⁸

Venus tidal acceleration / representative weather pressure-gradient acceleration

Venus's tidal acceleration is roughly 22 million times smaller.

This comparison is an order-of-magnitude illustration, not a direct calculation of a resulting weather pressure anomaly.

6. Proposed extension to your Bay of Bengal model

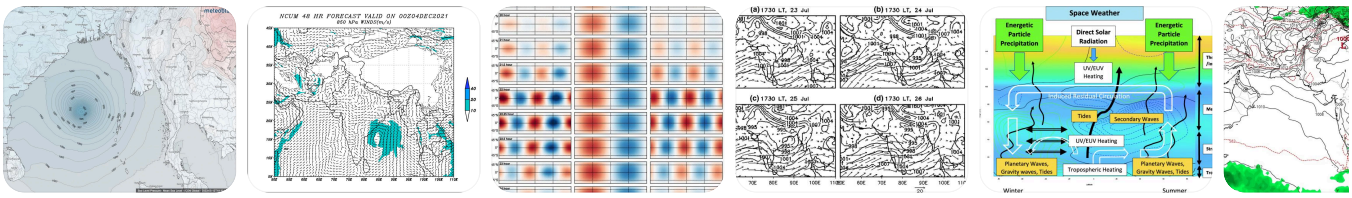
I recommend a controlled numerical experiment with four model configurations.

Model	Gravitational treatment
A	Standard Earth gravity
B	Earth + Sun + Moon tidal potential
C	Earth + Sun + Moon + Mercury + Venus
D	Full planetary N-body tidal potential

For each model, calculate:

- Daily mean sea-level pressure.
- Horizontal pressure gradients.
- Pressure-gradient force vectors.
- 850 hPa and 500 hPa geopotential heights.
- Wind velocity and vorticity.
- Vertical velocity.
- Precipitation and convective activity.

The difference between C and B would isolate the Mercury–Venus contribution. D would quantify the additional influence of other planets.



7. My scientific recommendation

I would not simply insert the full N-body acceleration into a weather model. I would:

1. Calculate planetary positions using JPL DE440/DE441 ephemerides.
2. Transform gravitational accelerations into the Earth-centred reference frame.
3. Calculate the external tidal potential and its spatial derivatives.
4. Add the resulting forcing to the atmospheric momentum equations.
5. Run paired simulations with identical initial conditions.
6. Compare pressure-gradient differences against ensemble forecast uncertainty.

This would establish whether any planetary signal is distinguishable from normal atmospheric variability.

The atmospheric modelling literature provides a sound foundation for this approach. AGU Journals +2

One particularly interesting research question: Can the time-varying planetary tidal potential produce a statistically detectable perturbation in Bay of Bengal atmospheric pressure, even if its direct acceleration is extremely small?

That is testable using controlled numerical experiments, although a detectable weather-scale effect should not be assumed.

I can develop this as a **Mercury–Venus–Earth N-body tidal forcing module for the Bay of Bengal Navier–Stokes model**, covering the daily period through 31 December 2026.

You said:

Yes

Analysis paused

ChatGPT said:

Bay of Bengal planetary tidal-forcing module

13 November – 31 December 2026 · Research prototype

I have developed the first computational module for the Bay of Bengal domain, using the previously established 5°N–22°N and 78°E–100°E boundaries.

It calculates the spatially varying gravitational acceleration due to Mercury and Venus, after subtracting the acceleration of Earth's centre of mass.

1. Computational architecture

```
JPL planetary orbital elements
          |
          v
Heliocentric planetary positions
          |
          v
Earth-centred relative coordinates
          |
          v
Earth-fixed coordinate transformation
          |
          v
Differential gravitational acceleration
          |
          v
```

```
Mercury + Venus tidal forcing
          |
          v
Bay of Bengal spatial grid
          |
          v
Horizontal and vertical forcing fields
          |
          v
Atmospheric momentum equation
          |
          v
Pressure-gradient response
```

The gravitational forcing at each grid point is calculated using:

$$\mathbf{a}_i(\mathbf{r}) = GM_i \left[\frac{\mathbf{R}_i - \mathbf{r}}{|\mathbf{R}_i - \mathbf{r}|^3} - \frac{\mathbf{R}_i}{|\mathbf{R}_i|^3} \right]$$

This subtracts Earth's centre-of-mass acceleration and retains the actual differential tidal field.

2. Computed Bay of Bengal results

0.5° grid spacing · Maximum horizontal tidal acceleration

Date	Mercury + Venus (m/s²)	Equivalent pressure-gradient forcing (Pa/m)
13 Nov	2.769×10^{-11}	3.323×10^{-11}
7 Dec	9.062×10^{-12}	1.087×10^{-11}
31 Dec	3.443×10^{-12}	4.132×10^{-12}

The equivalent pressure-gradient forcing uses an illustrative air density of 1.2 kg/m³:

$$\nabla p_{\text{equivalent}} = \rho \mathbf{a}_{\text{tidal,h}}$$

This represents the forcing in the momentum equation, **not an actual simulated atmospheric pressure gradient**.

Mercury–Venus tidal forcing over the Bay of Bengal

3. Comparison with the other gravitational sources

I also computed a comparison case containing the Sun and all eight planets, excluding the Moon.

Date	Mercury + Venus	Sun + eight planets

Maximum horizontal differential acceleration; selected dates, approximate orbital model.

13 Nov	2.77×10^{-11}	1.77×10^{-7}
7 Dec	9.06×10^{-12}	2.15×10^{-7}
31 Dec	3.44×10^{-12}	2.47×10^{-7}

Units: maximum horizontal acceleration in m/s².

This demonstrates why the Sun and Moon must be included in the complete atmospheric tidal model. Mercury and Venus are a very small additional perturbation.

4. Generated research files

Python computational module
Orbital positions and tidal-force calculations

Download Python module ↗

Daily forcing dataset
49 dates, two gravitational cases

Download daily summary CSV ↗

Spatial tidal grid
4,725 grid-point records across three dates

Download spatial grid CSV ↗

Methodology and limitations
Computational assumptions and validation requirements

Download methodology notes ↗

5. Scientific limitations and next development stage

The current version is a forcing-field prototype, not a completed weather simulation.

There are three important upgrades needed:

1. **Ephemeris accuracy:** Replace approximate Keplerian positions with JPL DE440 Cartesian state vectors. JPL explicitly distinguishes its approximate orbital formulae from its higher-precision ephemerides. JPL Solar Sys… +1
2. **Complete tidal forcing:** Include the Moon, Sun, and all planetary contributions, with accurate Earth orientation.
3. **Atmospheric integration:** Couple the forcing to the primitive atmospheric equations, including continuity, thermodynamics, moisture, Coriolis acceleration and boundary conditions.

The next experiment should compare two otherwise identical atmospheric simulations:

- Baseline: conventional solar and lunar tidal forcing.
- Perturbed: baseline plus Mercury and Venus tidal forcing.

The resulting difference in MSLP, wind and precipitation would test whether the planetary perturbation produces any detectable atmospheric response.

One important distinction: A nonzero gravitational perturbation is physically real, but it does not automatically imply a measurable weather effect.

6. Suggested next experiment

I recommend adding the Moon and producing a complete Sun–Moon–Mercury–Venus tidal forcing model before attempting the atmospheric simulation.

Would you like me to refresh the Bay of Bengal forcing calculations daily through 31 December?

Refresh Bay of Bengal tidal model daily
Tap to set up

‹/›

bob_planetary_tidal_forcing_module_approx.py
Code

bob_planetary_tidal_forcing_daily_summary_2026.csv
Spreadsheet

bob_mercury_venus_tidal_grid_snapshots_2026.csv
Spreadsheet

README_bob_planetary_tides_2026.txt
Document